

N4 Rocket Recovery System

Continuous Assessment Presentation

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March 2026

Nakuja Project — Final Year Project

Outline

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Introduction & Background

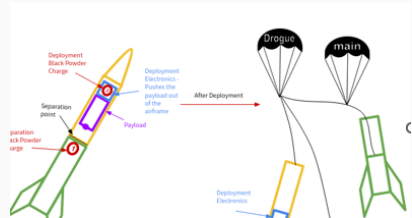
Context:

- The Nakuja Project is JKUAT's student rocketry programme, targeting altitude records with successive N-series rockets.
- The N4 rocket uses a **dual-deployment recovery system** â€” a drogue parachute at apogee stabilises descent; a main parachute deploys at low altitude for safe landing.
- Recovery depends on three tightly coupled systems: **telemetry** (track the rocket after separation), **pyrotechnic deployment** (eject the parachutes reliably) and **pre-flight monitoring**

N-3.5 rocket (2024)



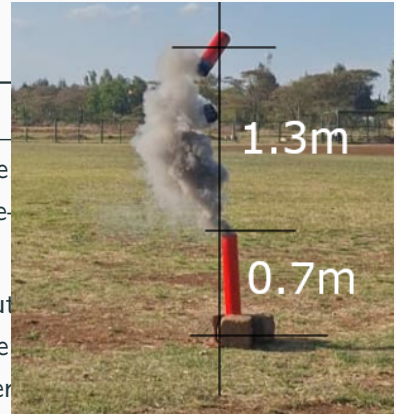
Nakuja N3 rocket â€” dual-deployment configuration



Problem Statement

Three failure modes identified from N3 field experience:

Problem Area	Consequence
Short telemetry range	Signal lost after apogee rocket unrecoverable beyond ≈ 1 km
DC pyro dump (no control)	Nichrome wire burns out in ~ 2 s; reuse impossible thermal damage to holder
No pre-flight diagnostics	Battery state and igniter continuity unknown at launch; deployment failure undetectable



Excessive crimson powder deployment from uncontrolled charge



Objectives

Main Objective

To design, fabricate, and validate a reliable dual-deployment recovery system for the N4 rocket, achieving telemetry coverage beyond 5 km, controlled pyrotechnic deployment, and real-time pre-flight system diagnostics.

Specific Objectives:

1. **Telemetry range:** Design and configure a 900 MHz XBee-PRO 900HP radio link with a verified link margin ≥ 20 dB at 5 km range to ensure post-apogee rocket tracking and telemetry reception.
2. **Controlled deployment:** Implement a PWM-based ignition controller driving an IRLZ44N MOSFET at a calibrated duty cycle to eliminate nichrome burnout and provide safe, repeatable pyrotechnic ejection charge activation.
3. **Structural design & in-flight monitoring:** Design a 3D-printed PETG avionics holder satisfying the 95 mm bay envelope constraint; integrate an ADS1115 16-bit

Literature Review â€” Telemetry, Deployment & Monitoring

Source	Key Contribution	Limitation / Gap
Digi International (2019) â€” XBee-PRO 900HP User Guide	900 MHz FHSS, +24 dBm Tx, â€“110 dBm Rx sensitivity; transparent UART mode for direct microcontroller integration	No application-level link-budget study for rocket attitude variation
Pozar (2011); ARRL Antenna Book (2019)	Free-space path loss model; dipole radiation pattern â€” omnidirectional main lobe with axis nulls	Null zones along dipole axis require tilt margin allocation
Nakka, R. (2021) â€” Pyrotechnic igniter design	PWM duty-cycle control limits peak current and prevents nichrome burnout; repeatability validated	Study uses bench test only; no in-rocket deployment validation
Martinez & Chen (2018) â€” PWM for resistive heating	Confirmed PWM at 1 kHz as effective thermal control for nichrome wire	Focus on industrial heaters; not validated for pyrotechnic ignition pulses

Literature Review – Structures, Firmware & Recovery

Source	Key Contribution	Limitation / Gap
Santana & Alves (2021) – PETG FDM properties	PETG tensile strength ≈ 50 MPa, chemical resistance, low warping vs PLA; viable for avionics enclosures	High-shock load data (>20 g) absent; vibration fatigue not characterised
Real Time Engineers Ltd (2023) – FreeRTOS Manual	Task scheduling, priority inversion, and inter-task communication for embedded systems	No rocket-specific FSM reference architecture provided
Knacke, T.W. (1991) – Parachute Recovery Systems	Drogue + main dual-deployment principles; deployment force and timing criteria	Pre-digital; no embedded trigger logic or apogee-detection algorithms discussed
Rodriguez, C. (2020) – MOS-FET Gate Drive	IRLZ44N logic-level enhancement at 3.3 V; $R_{DS(on)} = 22$ m Ω ; 40 μ s turn-on	Switching characterisation at low-side solenoid load; ignition-specific testing

Gaps Identified & Proposed Approach

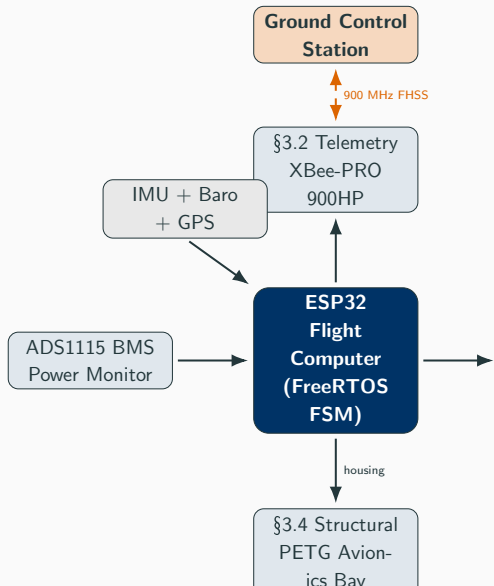
Gaps in existing solutions:

- No verified 5 km link budget for N4 attitude uncertainty â€™ literature gives theory only; no field-proven margin for this specific flight profile.
- DC pyrotechnic dump is single-use and thermally destructive â€™ no study combines PWM ignition with a logic-level MOSFET on an ESP32 platform.
- No integrated launch-inhibit system combining battery voltage monitoring *and* continuity detection on a single flight computer

Proposed approach:

1. **XBee-PRO 900HP** â€™ 900 MHz, +24 dBm; calculated link margin +32 dB at 5 km.
2. **PWM ignition** at 1 kHz, $D = 0.40$; IRLZ44N MOSFET driven directly from 3.3 V ESP32 GPIO.
3. **ADS1115 I²C ADC** â€™ 47 k Ω /10 k Ω divider; launch inhibit if $V_{bat} < 14.0$ V or continuity absent.
4. **Stepped PETG battery carrier** â€™ raised edge cells clear threaded rods; dual-rod retention for

System Architecture & Subsystem Overview

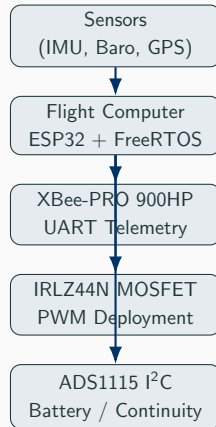


Subsystem	Key Hardware	Function
Telemetry (§3.2)	XBee-PRO 900HP; duck antenna	900 MHz link; +32 dB margin at 5 km; 3-channel comm manager
Electrical + Deploy (§3.3)	IRLZ44N MOSFET; ADS1115; custom PCB	PWM ignition ($D=0.40$); battery voltage + continuity

Methodology – Design Approach

4-module design structure:

#	Module	Purpose
1	Telemetry	Long-range rocket tracking & data down-link post-apogee
2	Pyrotechnic	Controlled ejection charge activation for dual deployment
3	Power Monitor	Pre-flight safety: battery state + igniter continuity
4	Mechanical	Structural containment within 95 mm low vibration isolation



System architecture overview

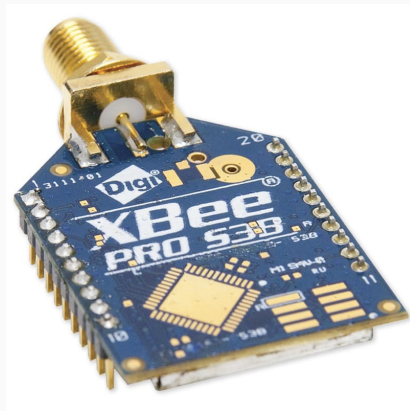
Methodology â€” Telemetry Module

Design Considerations:

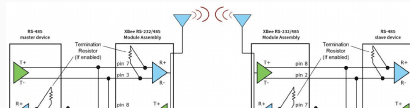
- Must maintain link over 5 km with rocket tumbling (attitude variation $\pm 60^\circ$)
- Must survive 20 g launch shock without connector failure
- Direct UART to ESP32 (no packet overhead)

Options evaluated:

Option	Freq.	Margin@5km
LoRa SX1276	433 MHz	+8 dB
433 MHz OOK	433 MHz	-2 dB (fail)
XBee-PRO 900HP	900 MHz	+32 dB



XBee-PRO 900HP module



Methodology â€” Pyrotechnic Deployment Module

Design Considerations:

- Must fire reliably, but *not* burn out the igniter
- Logic-level control from 3.3 V ESP32 GPIO
- Inductive flyback protection for solenoid-type loads

DC dump vs PWM trade study:

Criterion	DC Dump	PWM
Wire re-use	No	Yes
Current control	None	✓
Holder damage	High risk	Mitigated



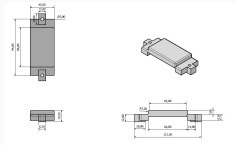
Cut nichrome wire after DC-dump burn-out
— single use



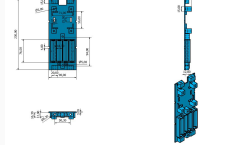
Methodology — Mechanical Module: CAD Designs

Design considerations: 95 mm avionics bay envelope; four 18650 cells parallel; threaded-rod retention paths; printable in PETG; upper platform for XBee and beacon.

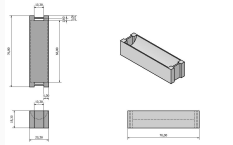
2D Engineering Drawings (dimensioned):



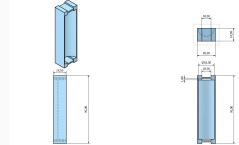
18650 case



FC holder sled

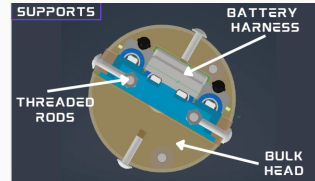
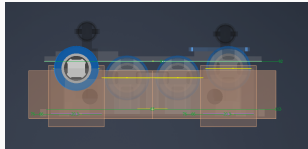
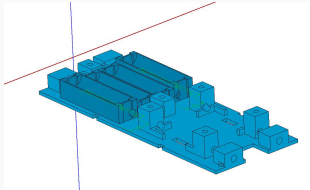


Edge carrier (raised)



Mid carrier

3D CAD Models & Assembly:

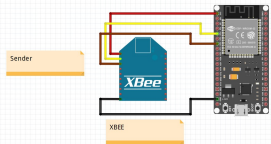


Design considerations:

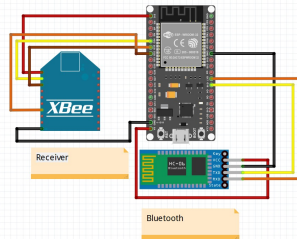
- Segregated pyro / logic power rails
- UART (115200 bps) for XBee; I²C for ADS1115
- All channels fused; MOSFET gate pull-down to prevent false fire

Development workflow:

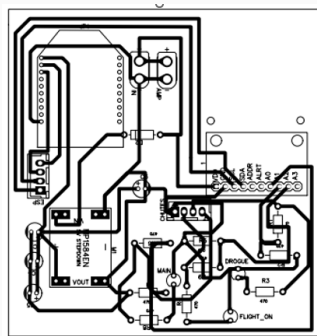
1. Breadboard validation
2. Fritzing integration layout
3. EasyEDA schematic + PCB
4. Proteus BMS simulation



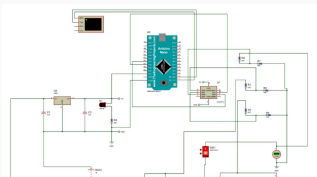
Fritzing "rocket wiring"



Fritzing " base station



EasyEDA schematic



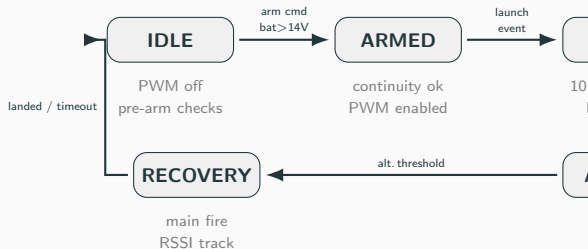
Methodology â€™ Firmware & Software Architecture

FreeRTOS task structure:

Task	Rate
Sensor fusion (IMU, baro)	10 Hz
XBee UART telemetry TX	10 Hz
FSM state machine	20 Hz
ADS1115 voltage monitor	1 Hz
Deployment (event-driven)	ISR

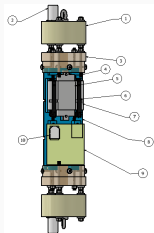
Deployment arm conditions:

- Battery >14.0 V AND continuity >1000 counts
- FSM state = APOGEE or RECOVERY
- Inhibit flag clear



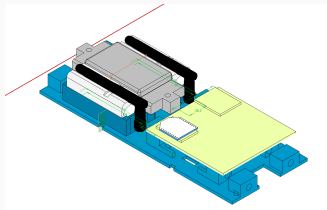
N4 firmware state machine

Results â€™ Mechanical Design Outputs

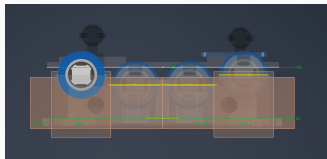


PARTS LIST			
PIN	QTY	AVIAT NUMBER	DESCRIPTION
1	2	1000000	Capacitor
2	2	1000000	Capacitor
3	2	1000000	Capacitor
4	2	1000000	Capacitor
5	2	1000000	Capacitor
6	2	1000000	Capacitor
7	2	1000000	Capacitor
8	2	1000000	Capacitor
9	2	1000000	Capacitor
10	2	1000000	Capacitor

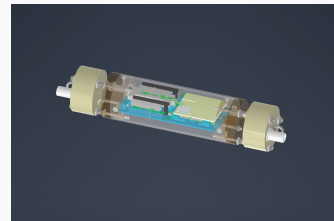
Full avionics bay parts list



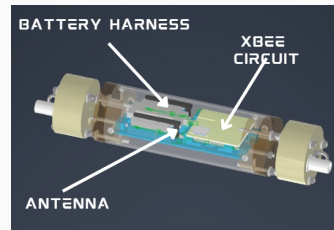
Assembled avionics bay



Rod clearance confirmed

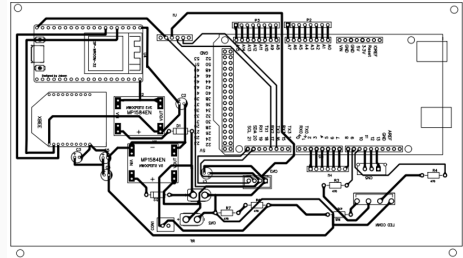
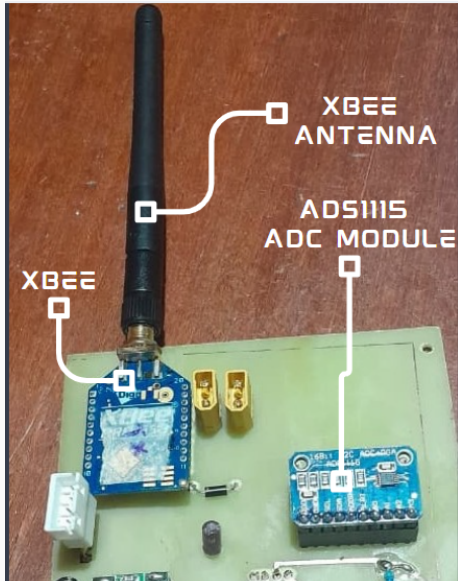


Full integration in airframe

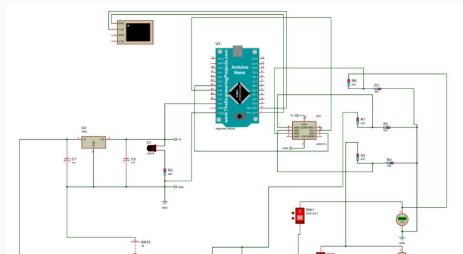


3D assembly / isometric view 16

Results â€™ Electronics & Avionics



Base station EasyEDA schematic



Budget

Item			Qty	Est. Cost (KES)
XBee-PRO 900HP modules			2	12,000
($\tilde{\text{A}}$ —2)				
ESP32 DevKit (flight + GCS)			2	1,200
PCB fabrication (chemical etch)			1	800
IRLZ44N MOSFETs + passives			—	400
ADS1115 ADC module			1	300
18650 battery cells ($\tilde{\text{A}}$ —4)			4	2,400
PETG filament (structural parts)			—	1,500
900 MHz duck antennas ($\tilde{\text{A}}$ —2)			2	800
Nichrome wire + connectors			—	600
Pop-test black powder charges			—	1,500
Miscellaneous (solder, PCB stock)			—	1,000
Total				22,500

All costs in Kenyan Shillings (KES). Procurement partially complete.

Conclusion & Way Forward

Progress summary:

- XBee-PRO 900HP configured; link budget verified (+32 dB at 5 km)
- PWM ignition validated: controlled firing without burnout at $D = 0.40$
- ADS1115 Proteus simulation confirms distinct ADC codes for launch-inhibit logic
- PETG structural CAD complete; stepped battery carrier resolves rod interference
- Custom PCB chemically etched and soldered (88 mm $\times$ 93 mm)
- FreeRTOS firmware modules bench-integrated; FSM tested

Objective Progress:

Objective	Status	%
Telemetry range	In progress	70%
PWM deployment	In progress	65%
BMS / structural	In progress	60%
System integration	Planned	0%

Time Plan

Task	N			D		J		F		M		A		M		J		
Proposal presentation																		
Continuous presentation																		
Literature review																		
FC Holder design																		
Xbee Circuit Design																		
Software																		
Interim report																		
Material requisition																		
PCB Fabrication																		
Range & Drone Test																		

Thank you

Questions?